DENSEST PACKINGS OF TWO TRANSLATES OF A LATTICE IN FOUR DIMENSIONS

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ABSTRACT. Let Λ be a lattice in $\mathbb{R}^4$ and b a vector outside Λ such that any two distinct points of $\Lambda \cup (\Lambda + b)$ are at distance at least 2. We prove that the covolume of Λ is at least 16, with equality only when $\Lambda \cup (\Lambda + b)$ is congruent to $\sqrt{2}\, D_4$. Thus no packing of unit balls whose centres form two translates of a lattice in $\mathbb{R}^4$ is denser than the D_4 lattice packing, and every Voronoi cell of such a packing has volume at least 8, the volume of the regular 24-cell, as Musin's 24-cell conjecture predicts. The packing condition on the Gram matrix of a basis of Λ together with b cuts out a locally finite polyhedron. The concavity of $\log\det$ and of a Schur complement reduces the problem to the vertices and edges of this polyhedron. Up to integral symmetries it has ten vertices and 1137 edges at them. This finite table is computed in exact arithmetic and re-checked by four independent programs; every other step is proved by hand.

1. INTRODUCTION

The root lattice $D_4 = \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \in 2\mathbb{Z}\}$, scaled by $\sqrt{2}$ so that its minimal distance is 2, carries a packing of unit balls of density $\pi^2/16$. Korkine and Zolotarev proved that no lattice packing of balls in $\mathbb{R}^4$ is denser [5]. Whether D_4 is densest among all packings in $\mathbb{R}^4$ is open. Musin's 24-cell conjecture [6] states that every Voronoi cell of a packing of unit balls in $\mathbb{R}^4$ has volume at least 8, the volume of the regular 24-cell, which is the Voronoi cell of $\sqrt{2}\, D_4$; it implies that D_4 is densest.

A periodic packing is a finite union of translates of one lattice. After lattices, the next case is two translates, $P = \Lambda \cup (\Lambda + b)$. Andreanov and Kallus [1] extended Voronoi's algorithm for perfect forms [8] to such sets and enumerated the locally optimal ones in dimensions 3, 4 and 5. In dimension 4 their enumeration found D_4 densest. Parts of their computation used floating-point arithmetic, and they present these results as numerical rather than rigorous [1, §4]. We prove the four-dimensional statement.

Theorem 1.1. *Let $\Lambda \subset \mathbb{R}^4$ be a lattice and $b \in \mathbb{R}^4 \setminus \Lambda$ such that $|x - y| \geq 2$ for all distinct $x, y \in P = \Lambda \cup (\Lambda + b)$. Then $\operatorname{covol}(\Lambda) \geq 16$, with equality if and only if P is congruent to $\sqrt{2}\, D_4$. Equivalently, the packing of unit balls centred at the points of P has density at most $\pi^2/16$, with equality only for $\sqrt{2}\, D_4$.*

Corollary 1.2. *In a packing of unit balls in $\mathbb{R}^4$ whose centres form two translates of a lattice, every Voronoi cell has volume at least 8. Equality holds only when the centres form $\sqrt{2}\, D_4$, and then every cell is a regular 24-cell.*

For a single lattice the bound of Corollary 1.2 is the theorem of Korkine and Zolotarev, so Corollary 1.2 proves the 24-cell conjecture for all packings whose centres form at most two translates of a lattice.

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Method. A two-periodic set is encoded by the 5×5 Gram matrix J of a basis of Λ together with b. The packing condition becomes infinitely many linear inequalities $J[k] \geq 4$, which cut out a locally finite polyhedron $\mathcal{R}$ (Section 2); packings are the points of $\mathcal{R}$ where $\det J = 0$, and the quantity to minimise is $\det Q$, where Q is the upper left 4×4 block of J. Proposition 3.2 shows that every packing can be moved, without increasing $\det Q$, to a vertex of $\mathcal{R}$ or to a point of an edge of $\mathcal{R}$. The argument uses only the concavity of $\log\det Q$ and of the Schur complement of Q in J; it replaces the analysis of local optimality conditions and of "fluid" packings in [1, Theorem 3].

Up to the integral symmetries of the problem, $\mathcal{R}$ has ten vertices and 1137 edges at them (Proposition 5.1). This is the one statement of the paper that rests on computation, which is finite and exact, and we give certificates that four independent programs re-check (Appendix A).[1] Everything else is proved by hand. The concavity of the Schur complement shows that bounded edges contain no packings in their interiors, and on the 25 unbounded edges all quantities are explicit (Table 5.3). The equality case uses a covering property of two sublattices of index two in D_4 (Lemma 4.2).

The method is specific to two translates. Three or more translates, and the 24-cell conjecture for arbitrary packings, are not treated here.

2. Two-periodic sets and the polyhedron $\mathcal{R}$

2.1. **Notation.** Sym_n is the space of real symmetric $n \times n$ matrices, with any fixed norm $\|\cdot\|$. For $J \in \mathrm{Sym}_5$ and $x \in \mathbb{R}^5$ we write $J[x] = x^{\mathsf{T}} J x$, and we write

$$J = \begin{pmatrix} Q & r \\ r^{\mathsf{T}} & s \end{pmatrix}, \qquad Q = Q(J) \in \mathrm{Sym}_4,\ r \in \mathbb{R}^4,\ s \in \mathbb{R}.$$

Let $\mathcal{M} = \mathbb{Z}^4 \times \{-1, 0, 1\} \subset \mathbb{Z}^5$ and $\mathcal{M}^* = \mathcal{M} \setminus \{0\}$. Elements of $\mathcal{M}$ are written $k = (n, l)$ with $n \in \mathbb{Z}^4$, $l \in \{-1, 0, 1\}$. Define

$$\mathcal{R} = \{J \in \mathrm{Sym}_5 :\ J[k] \geq 4 \text{ for all } k \in \mathcal{M}^*\}.$$

$\mathcal{R}$ is closed and convex. For $J \in \mathcal{R}$ let

$$\mathrm{Min}(J) = \{k \in \mathcal{M}^* : J[k] = 4\}, \qquad C_J = \{X \in \mathrm{Sym}_5 : X[k] \geq 0 \text{ for all } k \in \mathrm{Min}(J)\},$$
$$L_J = C_J \cap (-C_J),$$

so $L_J = \{X : X[k] = 0\ \forall k \in \mathrm{Min}(J)\}$, and let $d(J) = \dim L_J$. Since k and $-k$ give the same condition, we list one vector of each pair $\pm k$ when counting $\mathrm{Min}(J)$.

Let $\Lambda \subset \mathbb{R}^4$ be a lattice with basis $a_1, \dots, a_4$ and $b \in \mathbb{R}^4$. The Gram matrix of $(a_1, \dots, a_4, b)$ is denoted $J(a, b)$. Then

$$J(a,b)[(n,l)] = \Big|\sum_{i=1}^{4} n_i a_i + lb\Big|^2. \tag{2.1}$$

Lemma 2.1. *Let $b \notin \Lambda$. The distinct points of $\Lambda \cup (\Lambda + b)$ are at distance at least 2 if and only if $J(a, b) \in \mathcal{R}$.*

Proof. Since Λ and $\Lambda + b$ are disjoint, the differences of distinct points are the vectors of $(\Lambda \setminus 0) \cup (\Lambda + b) \cup (\Lambda - b)$, that is, the vectors $\sum n_i a_i + lb$ with $(n, l) \in \mathcal{M}^*$. Apply (2.1). □

Lemma 2.2. *For $J \in \mathcal{R}$ the block $Q(J)$ is positive definite.*

[1] The ten vertices agree with those found numerically by Andreanov and Kallus [1].

Proof. We have $Q[n] = J[(n,0)] \geq 4$ for $n \in \mathbb{Z}^4 \setminus 0$. If $Q[x] < 0$ for some $x \in \mathbb{R}^4$, the open cone $\{Q < 0\}$ contains a rational point and hence an integral point $n \neq 0$ with $Q[n] < 0$, a contradiction. So Q is positive semidefinite. Suppose $Qv = 0$ with $\max_i |v_i| = 1$. By Dirichlet's theorem on simultaneous approximation, for every integer $N \geq 1$ there are an integer $1 \leq m \leq N^4$ and $n \in \mathbb{Z}^4$ with $\max_i |mv_i - n_i| \leq 1/N$. Then $n \neq 0$ because $\max_i |mv_i| = m \geq 1 > 1/N$, and $Q[n] = Q[n - mv] \leq 16\|Q\|_\infty/N^2$, where $\|Q\|_\infty$ is the largest absolute value of an entry. For large N this is less than 4, a contradiction. Hence Q is positive definite. □

Let $\mathrm{Sym}_5^+ = \{J \in \mathrm{Sym}_5 : Q(J) \text{ positive definite}\}$, an open convex set containing $\mathcal{R}$. For $J \in \mathrm{Sym}_5^+$ put

$$q(J) = Q^{-1}r, \qquad \sigma(J) = s - r^{\mathsf{T}}Q^{-1}r.$$

Lemma 2.3. *Let $J \in \mathrm{Sym}_5^+$.*

1. *$J[(x,l)] = Q[x + l\,q(J)] + l^2\sigma(J)$ for $x \in \mathbb{R}^4$ and $l \in \mathbb{R}$.*
2. *$\sigma(J) = \min_{x\in\mathbb{R}^4} J[(x,1)]$, attained at $x = -q(J)$. In particular $\sigma(J') \leq J'[u]$ for every $J' \in \mathrm{Sym}_5^+$ and every $u \in \mathbb{R}^4 \times \{1\}$, and σ is concave on Sym_5^+.*
3. *$\det J = \det Q \cdot \sigma(J)$.*

Proof. (1) is completing the square, (2) follows from (1) because Q is positive definite, and a minimum of affine functions of J is concave. (3) is the Schur complement formula. □

Lemma 2.4. *Let $J \in \mathcal{R}$.*

1. *$\sigma(J) = 0$ if and only if $J = J(a,b)$ for a basis a of a lattice Λ and a vector $b \notin \Lambda$ such that the distinct points of $\Lambda \cup (\Lambda + b)$ are at distance at least 2. In that case $\det Q(J) = \mathrm{covol}(\Lambda)^2$.*
2. *If $\sigma(J) < 0$, then $J + |\sigma(J)|\, e_5 e_5^{\mathsf{T}} \in \mathcal{R}$, its Q-block is $Q(J)$, and its σ is 0.*

Proof. (1) If $\sigma(J) = 0$, then J is positive semidefinite of rank 4 by Lemma 2.3(1), (3) and Lemma 2.2, hence the Gram matrix of five vectors $a_1, \dots, a_4, b$ of $\mathbb{R}^4$, and $a_1, \dots, a_4$ are linearly independent because Q is positive definite. If b were $\sum n_i a_i$, then $J[(-n,1)] = 0 < 4$. Lemma 2.1 gives the packing property. Conversely $J(a,b)$ is a Gram matrix of five vectors in $\mathbb{R}^4$, so $\det J(a,b) = 0$ and $\sigma = 0$ by Lemma 2.3(3). Finally $\det Q = \det(a_i \cdot a_j) = \mathrm{covol}(\Lambda)^2$. (2) Adding $|\sigma| e_5 e_5^{\mathsf{T}}$ changes $J[(n,l)]$ by $l^2|\sigma| \geq 0$, leaves Q and r unchanged and raises s, hence σ, by $|\sigma|$. □

Lemma 2.5. *For $J \in \mathcal{R}$, $\det Q(J) \geq \pi^4/4$.*

Proof. Q is the Gram matrix of a lattice Λ with no nonzero vector of length less than 2. The open ball of radius 2 in $\mathbb{R}^4$ has volume $8\pi^2$. By Minkowski's convex body theorem [3, Ch. III] it contains a nonzero point of Λ if $8\pi^2 > 2^4 \mathrm{covol}(\Lambda)$. Hence $\mathrm{covol}(\Lambda) \geq \pi^2/2$. □

2.2. **Local structure.**

Lemma 2.6.

1. *For every compact set $K \subset \mathrm{Sym}_5^+$ the set $\mathcal{M}_K = \{k \in \mathcal{M}^* : J[k] \leq 5 \text{ for some } J \in K\}$ is finite.*
2. *For every $J \in \mathcal{R}$ there is $\varepsilon > 0$ such that for all $X \in \mathrm{Sym}_5$ with $\|X\| \leq \varepsilon$ we have $J + X \in \mathcal{R}$ if and only if $X \in C_J$.*
3. *For every $J \in \mathcal{R}$, $\mathrm{Min}(J)$ is finite and $\mathcal{R} \subseteq J + C_J$.*

Proof. (1) On K the smallest eigenvalue of Q is at least some $\mu > 0$, and $|q(J)| \leq \rho$, $|\sigma(J)| \leq \rho$ for some ρ, by continuity. For $k = (n,l) \in \mathcal{M}^*$ and $J \in K$, Lemma 2.3(1) gives $J[k] \geq \mu|n + lq(J)|^2 - \rho$. If $|n| > \rho + \sqrt{(5+\rho)/\mu}$, then $|n + lq(J)| \geq |n| - \rho$ and $J[k] > 5$. (2) Choose $\varepsilon_0 > 0$ such that the closed ball $K = \bar{B}(J, \varepsilon_0)$ lies in Sym_5^+. The finitely many

$k \in \mathcal{M}_K \setminus \mathrm{Min}(J)$ have $J[k] > 4$, so there is $\varepsilon \le \varepsilon_0$ with $(J+X)[k] > 4$ for all of them whenever $\|X\| \le \varepsilon$. For such X, the vectors $k \notin \mathcal{M}_K$ satisfy $(J+X)[k] > 5$, so $J + X \in \mathcal{R}$ if and only if $(J+X)[k] = 4 + X[k] \ge 4$ for all $k \in \mathrm{Min}(J)$. (3) $\mathrm{Min}(J) \subseteq \mathcal{M}_{\{J\}}$ is finite. If $J' \in \mathcal{R}$, then $J + \lambda(J'-J) \in \mathcal{R}$ for $\lambda \in [0,1]$, and by (2) $\lambda(J'-J) \in C_J$ for small $\lambda > 0$; C_J is a cone. □

Lemma 2.7. *$\mathcal{R}$ contains no line.*

Proof. If $J + \tau X \in \mathcal{R}$ for all $\tau \in \mathbb{R}$, then $X[k] = 0$ for every $k \in \mathcal{M}^*$. The matrices kk^{T} for $k = e_i$ $(1 \le i \le 5)$ and $k = e_i + e_j$ $(1 \le i < j \le 5)$, all in $\mathcal{M}^*$, span Sym_5, so $X = 0$. □

Definition 2.8. A *vertex* of $\mathcal{R}$ is a point $J \in \mathcal{R}$ with $d(J) = 0$. Then C_J is a pointed polyhedral cone, defined by the finitely many inequalities $X[k] \ge 0$, $k \in \mathrm{Min}(J)$, whose linear forms span the 15-dimensional dual of Sym_5. An *extreme ray* of C_J is a ray $\mathbb{R}_{\ge 0}Y$, $Y \ne 0$, with $Y \in C_J$ such that $\{k \in \mathrm{Min}(J) : Y[k] = 0\}$ has rank 14, i.e. the forms $X \mapsto X[k]$ for these k span a 14-dimensional space. For an extreme ray Y put

$$t^*(J,Y) = \sup\{t \ge 0 : J + tY \in \mathcal{R}\} \in (0,\infty],$$

which is positive by Lemma 2.6(2). The set $\{J + tY : 0 \le t \le t^*\}$ (with $t < \infty$) is the *edge* of $\mathcal{R}$ at J in direction Y; it is *bounded* if $t^* < \infty$.

We use two standard facts on polyhedral cones [9, Lecture 1]: a pointed polyhedral cone has finitely many extreme rays and is their conical hull, and a nonzero vector of the cone spans an extreme ray exactly when the inequalities it satisfies with equality have rank one less than the dimension.

Lemma 2.9. *Let J be a vertex and Y an extreme ray of C_J.*

1. *$\{t \ge 0 : J + tY \in \mathcal{R}\}$ is the interval $[0, t^*]$, or $[0,\infty)$ if $t^* = \infty$. If $t^* < \infty$, then $J + t^*Y$ is a vertex.*
2. *Every $J' \in \mathcal{R}$ with $d(J') = 1$ lies on an edge: $J' = J_v + tY'$ for a vertex J_v, an extreme ray Y' of C_{J_v}, and $0 < t < t^*(J_v, Y')$.*

Proof. (1) The set is convex and closed. Let $J_1 = J + t^*Y$ with $t^* < \infty$ and $Z = \{k \in \mathrm{Min}(J) : Y[k] = 0\}$. Every $k \in Z$ has $J_1[k] = 4$, so $Z \subseteq \mathrm{Min}(J_1)$. Since $J_1 + \eta Y \notin \mathcal{R}$ for $\eta > 0$, Lemma 2.6(2) gives $Y \notin C_{J_1}$, so some $k_1 \in \mathrm{Min}(J_1)$ has $Y[k_1] < 0$. The forms of Z vanish on Y and span a 14-dimensional space, and the form of k_1 does not vanish on Y; together they have rank 15, so $L_{J_1} = 0$. (2) Let $L_{J'} = \mathbb{R}X_0$. The set $I_0 = \{\tau : J' + \tau X_0 \in \mathcal{R}\}$ is a closed interval which contains a neighbourhood of 0 by Lemma 2.6(2) and is not $\mathbb{R}$ by Lemma 2.7. Let τ_e be a finite end point, $J_v = J' + \tau_e X_0$ and $Y' = -\mathrm{sgn}(\tau_e)X_0$. Since $X_0[k] = 0$ for $k \in \mathrm{Min}(J')$, we have $\mathrm{Min}(J') \subseteq \mathrm{Min}(J_v)$ and $L_{J_v} \subseteq L_{J'}$. If $X_0 \in L_{J_v}$, then $J_v \pm \eta X_0 \in \mathcal{R}$ for small η by Lemma 2.6(2), contradicting the choice of τ_e; so $d(J_v) = 0$. The segment from J_v to J' lies in $\mathcal{R}$, so $Y' \in C_{J_v}$ by Lemma 2.6(2), and Y' vanishes on $\mathrm{Min}(J')$, which has rank 14; so Y' spans an extreme ray of C_{J_v}, and $J' = J_v + |\tau_e|Y'$. Finally $|\tau_e| < t^*(J_v, Y')$, since otherwise J' would be a vertex by (1). □

2.3. **Symmetries.** Let Γ be the group of matrices

$$T = \begin{pmatrix} U & v \\ 0 & \varepsilon \end{pmatrix}, \qquad U \in \mathrm{GL}_4(\mathbb{Z}),\ v \in \mathbb{Z}^4,\ \varepsilon \in \{1,-1\}.$$

Then $T(n,l) = (Un + lv, \varepsilon l)$, so T maps $\mathcal{M}$ bijectively onto itself, and $J \mapsto T^{\mathsf{T}}JT$ maps $\mathcal{R}$ bijectively onto itself with $(T^{\mathsf{T}}JT)[k] = J[Tk]$. Consequently $\mathrm{Min}(T^{\mathsf{T}}JT) = T^{-1}\,\mathrm{Min}(J)$, $X \mapsto T^{\mathsf{T}}XT$ maps C_J onto $C_{T^{\mathsf{T}}JT}$, and vertices, extreme rays, edges and the values t^* correspond.

Moreover $Q(T^{\mathsf T}JT) = U^{\mathsf T}QU$, so $\det Q$, $\det J$ and $\sigma = \det J/\det Q$ are Γ-invariant. In terms of point sets, T replaces the basis a by aU and b by $\varepsilon b + \sum v_i a_i$, so $J(a,b)$ and $T^{\mathsf T}J(a,b)T$ describe congruent sets. We call J, J' *equivalent* if $J' = T^{\mathsf T}JT$ for some $T \in \Gamma$.

2.4. **Reaching all vertices.**

Proposition 2.10. *Let v and w be vertices of $\mathcal R$. There are vertices $v = v_0, v_1, \dots, v_m = w$ such that $v_{s+1} = v_s + t^* Y_s$ for an extreme ray Y_s of C_{v_s} with $t^*(v_s, Y_s) < \infty$.*

Proof. Let $\ell(X) = \sum_{k\in \mathrm{Min}(w)} X[k]$. Then $\ell \ge 0$ on C_w, and $\ell(X) = 0$ for $X \in C_w$ only if $X \in L_w = 0$. By compactness of $\{X \in C_w : \|X\| = 1\}$ there is $\delta > 0$ with $\ell(X) \ge \delta\|X\|$ on C_w. By Lemma 2.6(3), $\mathcal R \subseteq w + C_w$, so $\ell(J) > \ell(w)$ for $J \in \mathcal R \setminus \{w\}$, and for every c the set $S_c = \{J \in \mathcal R : \ell(J) \le c\}$ is compact. A vertex J is the unique solution of the linear equations $J[k] = 4$, $k \in \mathrm{Min}(J)$, and the vertices in S_c have $\mathrm{Min}(J) \subseteq \mathcal M_{S_c}$, a finite set by Lemma 2.6(1) (note $S_c \subset \mathcal R \subset \mathrm{Sym}_5^+$). So S_c contains finitely many vertices.

Let u be a vertex with $u \ne w$. If $\ell(Y) \ge 0$ for every extreme ray of C_u, then $\ell \ge 0$ on C_u, and $\mathcal R \subseteq u + C_u$ gives $\ell(w) \ge \ell(u)$, which is false. So some extreme ray Y of C_u has $\ell(Y) < 0$. Along $u + tY$ the value of ℓ decreases, and $S_{\ell(u)}$ is compact, so $t^*(u, Y) < \infty$, and $u' = u + t^*Y$ is a vertex (Lemma 2.9) with $\ell(u') < \ell(u)$. Starting from $u = v$ and repeating, we obtain a sequence of vertices in $S_{\ell(v)}$ along which ℓ strictly decreases. It is finite, and it can only stop at w. □

Corollary 2.11. *Let $J_0, \dots, J_N$ be vertices of $\mathcal R$ such that for every i and every extreme ray Y of C_{J_i} with $t^*(J_i, Y) < \infty$ the vertex $J_i + t^*Y$ is equivalent to some J_j. Then every vertex of $\mathcal R$ is equivalent to some J_j, and every edge of $\mathcal R$ is equivalent to an edge at some J_j.*

Proof. Let w be a vertex and take the path $J_0 = v_0, \dots, v_m = w$ of Proposition 2.10. If $v_s = T^{\mathsf T}J_iT$, then $Y' = T^{-\mathsf T}Y_sT^{-1}$ is an extreme ray of C_{J_i} with the same t^*, and $v_{s+1} = T^{\mathsf T}(J_i + t^*Y')T$, which is equivalent to some J_j by hypothesis. By induction w is equivalent to some J_j. An edge at a vertex $w = T^{\mathsf T}J_jT$ in direction Y is the image under T of the edge at J_j in direction $T^{-\mathsf T}YT^{-1}$. □

3. Reduction to vertices and edges

Lemma 3.1. *Let Q be positive definite and $H \in \mathrm{Sym}_4$, $H \ne 0$. On the open interval of τ where $Q + \tau H$ is positive definite, $\psi(\tau) = \log\det(Q + \tau H)$ is strictly concave.*

Proof. With $Q_\tau = Q + \tau H$ we have $\psi'(\tau) = \mathrm{tr}(Q_\tau^{-1}H)$ and $\psi''(\tau) = -\mathrm{tr}((Q_\tau^{-1/2}HQ_\tau^{-1/2})^2) < 0$, because $Q_\tau^{-1/2}HQ_\tau^{-1/2}$ is a nonzero symmetric matrix. □

Let

$$S = \{J \in \mathcal R : \sigma(J) \le 0\}, \qquad \mathcal C = \{J \in S : d(J) = 0\} \ \cup\ \{J \in \mathcal R : d(J) = 1,\ \sigma(J) = 0\}.$$

So $\mathcal C$ consists of the vertices with $\det J \le 0$ and of the points with $\det J = 0$ in the relative interiors of edges. By Lemma 2.4, the infimum of $\det Q$ over S equals the infimum of $\mathrm{covol}(\Lambda)^2$ over the sets of Theorem 1.1.

Proposition 3.2. *For every $J \in S$ there is $J' \in \mathcal C$ with $\det Q(J') \le \det Q(J)$. If $\det Q(J) = \inf_S \det Q$, then J' can be chosen with $Q(J') = Q(J)$.*

Proof. Write $f(J) = \log\det Q(J)$ and $\pi(J) = 2d(J) + [\sigma(J) < 0]$, where $[\cdot]$ is 1 if the condition holds and 0 otherwise. We show: if $J \in S \setminus \mathcal C$, there is $J_1 \in S$ with $f(J_1) \le f(J)$ and $\pi(J_1) < \pi(J)$, and $Q(J_1) = Q(J)$ if $f(J) = \inf_S f$. Induction on π then proves the proposition.

Let $J \in S \setminus \mathcal{C}$. Then either (a) $\sigma(J) < 0$ and $d(J) \geq 1$, or (b) $\sigma(J) = 0$ and $d(J) \geq 2$. In case (a) choose any $X \in L_J \setminus 0$. In case (b) let $u = (-q(J), 1) \in \mathbb{R}^5$, so that $J[u] = \sigma(J) = 0$ by Lemma 2.3(2), and choose $X \in L_J \setminus 0$ with $X[u] = 0$, which exists because $\dim L_J \geq 2$. Put $J(\tau) = J + \tau X$ and

$$I_0 = \{\tau \in \mathbb{R} : J(\tau) \in \mathcal{R}\}.$$

I_0 is a closed interval, it contains a neighbourhood of 0 by Lemma 2.6(2) because $\pm X \in C_J$, and $I_0 \neq \mathbb{R}$ by Lemma 2.7. Since $X[k] = 0$ for $k \in \mathrm{Min}(J)$, we have $J(\tau)[k] = 4$ for these k, hence

$$\mathrm{Min}(J) \subseteq \mathrm{Min}(J(\tau)) \quad \text{and} \quad L_{J(\tau)} \subseteq L_J \qquad (\tau \in I_0). \tag{3.1}$$

In case (b), Lemma 2.3(2) gives $\sigma(J(\tau)) \leq J(\tau)[u] = J[u] + \tau X[u] = 0$ for $\tau \in I_0$, and we put $I = I_0$. In case (a), $\tau \mapsto \sigma(J(\tau))$ is continuous on I_0 and negative at 0, and we let I be the connected component of $\{\tau \in I_0 : \sigma(J(\tau)) \leq 0\}$ containing 0. In both cases I is a closed interval, 0 is an interior point of I, $I \neq \mathbb{R}$, and $J(\tau) \in S$ for $\tau \in I$.

The function $\varphi(\tau) = f(J(\tau))$ is concave on I by Lemma 3.1 (it is constant if $Q(X) = 0$), and bounded below by Lemma 2.5. If $I = [\alpha, \beta]$ is bounded, concavity gives $\varphi(0) \geq \min(\varphi(\alpha), \varphi(\beta))$. If $I = [\alpha, \infty)$, then φ is nondecreasing, because a concave function that decreases somewhere on a half-line tends to $-\infty$; so $\varphi(\alpha) \leq \varphi(0)$, and similarly for $I = (-\infty, \beta]$. In every case there is a finite end point τ_0 of I with $\varphi(\tau_0) \leq \varphi(0)$. Put $J_1 = J(\tau_0)$. Then $J_1 \in S$ and $f(J_1) \leq f(J)$.

If τ_0 is an end point of I_0, then $X \notin L_{J_1}$: otherwise $J_1 \pm \eta X \in \mathcal{R}$ for small $\eta > 0$ by Lemma 2.6(2). With (3.1), $d(J_1) \leq d(J) - 1$ and $\pi(J_1) \leq 2d(J) - 1 < \pi(J)$. If τ_0 is not an end point of I_0, we are in case (a), $\sigma(J_1) = 0$ by the choice of I, and $\pi(J_1) = 2d(J_1) \leq 2d(J) < \pi(J)$.

Now let $f(J) = m := \inf_S f$. Then $\varphi \geq m$ on I and $\varphi(0) = m$. Let τ_1 lie strictly between 0 and τ_0, and let $\tau' \in I$ lie on the other side of 0. Writing $0 = \lambda\tau' + (1-\lambda)\tau_1$ with $0 < \lambda < 1$, concavity gives $m \geq \lambda\varphi(\tau') + (1-\lambda)\varphi(\tau_1) \geq \lambda m + (1-\lambda)\varphi(\tau_1)$, so $\varphi(\tau_1) \leq m$, hence $\varphi(\tau_1) = m$. Thus φ is constant on the nondegenerate interval between 0 and τ_0, Lemma 3.1 forces $Q(X) = 0$, and $Q(J_1) = Q(J)$. Also $f(J_1) = m$, so the induction carries the equality case along. □

4. Two sublattices of index two in D_4

For $n \geq 2$ let $D_n = \{x \in \mathbb{Z}^n : \sum x_i \in 2\mathbb{Z}\}$. In D_4 consider the sublattices of index two

$$\Lambda_1 = \{x \in D_4 : x_1 \in 2\mathbb{Z}\} = 2\mathbb{Z} \oplus D_3, \qquad \Lambda_2 = \{x \in D_4 : x_1 + x_2 \in 2\mathbb{Z}\} = D_2 \oplus D_2,$$

where $D_2 \oplus D_2$ refers to the coordinate pairs (x_1, x_2) and (x_3, x_4).

Lemma 4.1. *Let $n \in \{2, 3\}$. For every $z \in \mathbb{R}^n$, $\mathrm{dist}(z, D_n) \leq 1$, with equality if and only if $z \in \mathbb{Z}^n \setminus D_n$. For every $t \in \mathbb{R}$, $\mathrm{dist}(t, 2\mathbb{Z}) \leq 1$, with equality if and only if t is an odd integer.*

Proof. The second statement is clear. Let $m \in \mathbb{Z}^n$ with $|z_i - m_i| \leq \frac{1}{2}$ for all i. If $m \in D_n$, then $\mathrm{dist}(z, D_n)^2 \leq n/4 < 1$. Otherwise let $\delta = \max_i |z_i - m_i|$, attained at i, and let $m' = m + \epsilon e_i$ with $\epsilon = 1$ if $z_i \geq m_i$ and $\epsilon = -1$ otherwise. Then $m' \in D_n$ and

$$|z - m'|^2 = |z - m|^2 - \delta^2 + (1 - \delta)^2 \leq n\delta^2 - 2\delta + 1.$$

Since $0 \leq \delta \leq \frac{1}{2} < \frac{2}{n}$, this is at most 1, with equality only for $\delta = 0$, that is $z = m \in \mathbb{Z}^n \setminus D_n$. Conversely, for $z \in \mathbb{Z}^n \setminus D_n$ and $x \in D_n$ the vector $x - z$ is integral with odd coordinate sum, so $|x - z| \geq 1$. □

Lemma 4.2. *For $i \in \{1, 2\}$ and every $y \in \mathbb{R}^4$, $\mathrm{dist}(y, \Lambda_i) \leq \sqrt{2}$, with equality if and only if $y \in D_4 \setminus \Lambda_i$.*

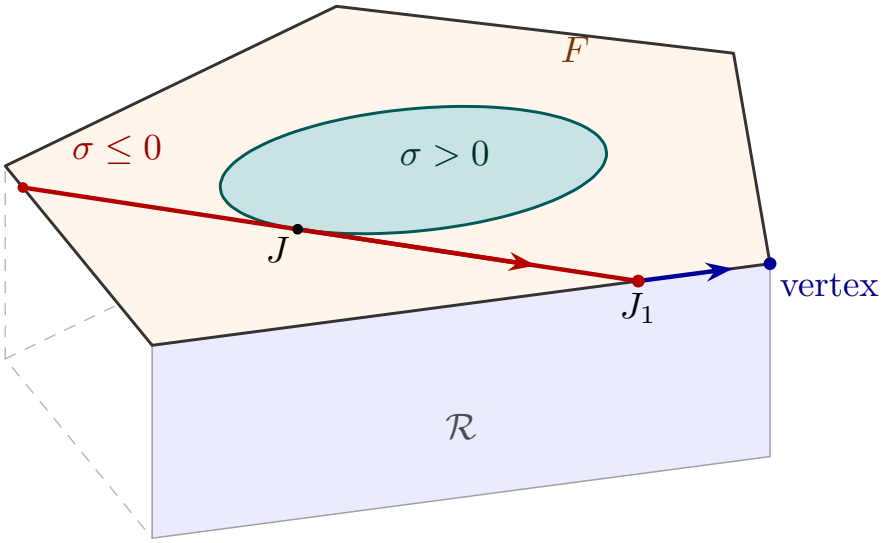


FIGURE 3.1. One step of the proof of Proposition 3.2 in a two-dimensional face F of $\mathcal{R}$, the case $\sigma(J) = 0$. The set $F \cap \{\sigma > 0\}$ (shaded) is convex because σ is concave. On the line through J in the direction X, chosen with $X[u] = 0$, we have $\sigma \le 0$, and $\log\det Q$ is concave on it, so one of the two ends of the chord, here J_1, has $\det Q(J_1) \le \det Q(J)$. The point J_1 lies on an edge of F. From there the same step moves along the edge to a vertex or to a point with $\sigma = 0$.

Proof. The sums are orthogonal. For Λ_1, $\operatorname{dist}(y,\Lambda_1)^2 = \operatorname{dist}(y_1, 2\mathbb{Z})^2 + \operatorname{dist}((y_2,y_3,y_4), D_3)^2 \le 2$, with equality if and only if y_1 is odd and $(y_2,y_3,y_4) \in \mathbb{Z}^3 \setminus D_3$ (Lemma 4.1), that is, $y \in \mathbb{Z}^4$, y_1 odd and $\sum y_i$ even, which means $y \in D_4 \setminus \Lambda_1$. For Λ_2, $\operatorname{dist}(y,\Lambda_2)^2 = \operatorname{dist}((y_1,y_2), D_2)^2 + \operatorname{dist}((y_3,y_4), D_2)^2 \le 2$, with equality if and only if $y \in \mathbb{Z}^4$ with $y_1 + y_2$ and $y_3 + y_4$ odd, which means $y \in D_4 \setminus \Lambda_2$. □

Corollary 4.3. *Let $\Lambda = \phi(\sqrt{2}\,\Lambda_i)$ for a linear isometry ϕ of $\mathbb{R}^4$ and $i \in \{1,2\}$. If $\operatorname{dist}(b,\Lambda) \ge 2$, then $\Lambda \cup (\Lambda + b) = \phi(\sqrt{2}\,D_4)$.*

Proof. $y = \phi^{-1}(b)/\sqrt{2}$ has $\operatorname{dist}(y,\Lambda_i) \ge \sqrt{2}$, so $y \in D_4 \setminus \Lambda_i$ by Lemma 4.2. As $[D_4 : \Lambda_i] = 2$, $D_4 = \Lambda_i \cup (\Lambda_i + y)$; apply $\sqrt{2}\,\phi$. □

The lattices $\sqrt{2}\,\Lambda_2$ and $\sqrt{2}\,\Lambda_1$ have the Gram matrices

$$G_2 = 4I_4 \quad (\text{basis } \sqrt{2}(e_1 \pm e_2),\ \sqrt{2}(e_3 \pm e_4)), \qquad G_1 = \begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & 4 & 2 & 2 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 4 \end{pmatrix} \tag{4.1}$$

(basis $2\sqrt{2}\,e_1$, $\sqrt{2}(e_2+e_3)$, $\sqrt{2}(e_2+e_4)$, $\sqrt{2}(e_3+e_4)$; here e_2+e_3, e_2+e_4, e_3+e_4 lie in D_3 and have determinant $\pm 2 = [\mathbb{Z}^3 : D_3]$, so they form a basis of D_3). Both have determinant 256.

5. THE VERTICES AND EDGES OF $\mathcal{R}$

Proposition 5.1. *The matrices $J_0, \dots, J_9$ of Table 5.1 are vertices of $\mathcal{R}$ with the stated invariants, and their edges are as in Table 5.2. In particular:*

i	J_i	$\lvert\operatorname{Min}(J_i)\rvert$	$\det Q$	$\det J$	$\lvert\Gamma_{J_i}\rvert$	rays	orbits
0	$\begin{pmatrix} 4 & 0 & 0 & 0 & 2 \\ 0 & 4 & 0 & 0 & -2 \\ 0 & 0 & 4 & 0 & -2 \\ 0 & 0 & 0 & 4 & -2 \\ 2 & -2 & -2 & -2 & 4 \end{pmatrix}$	20	256	0	768	60	4
1	$\begin{pmatrix} 4 & 0 & 0 & 0 & 2 \\ 0 & 4 & 2 & 2 & -4 \\ 0 & 2 & 4 & 2 & -4 \\ 0 & 2 & 2 & 4 & -4 \\ 2 & -4 & -4 & -4 & 8 \end{pmatrix}$	19	128	128	192	169	11
2	$\begin{pmatrix} 4 & 0 & 0 & 0 & 2 \\ 0 & 4 & 0 & 0 & -2 \\ 0 & 0 & 4 & 2 & -2 \\ 0 & 0 & 2 & 4 & -2 \\ 2 & -2 & -2 & -2 & 4 \end{pmatrix}$	17	192	128	96	33	5
3	$\begin{pmatrix} 4 & -2 & -2 & -2 & 6 \\ -2 & 4 & 2 & 2 & -6 \\ -2 & 2 & 4 & 2 & -6 \\ -2 & 2 & 2 & 4 & -6 \\ 6 & -6 & -6 & -6 & 16 \end{pmatrix}$	20	80	128	240	400	16
4	$\begin{pmatrix} 4 & 0 & -2 & 2 & 2 \\ 0 & 4 & 2 & 2 & -4 \\ -2 & 2 & 4 & 0 & -4 \\ 2 & 2 & 0 & 4 & -2 \\ 2 & -4 & -4 & -2 & 8 \end{pmatrix}$	20	64	128	768	400	9
5	$\begin{pmatrix} 4 & 0 & -1 & 1 & 2 \\ 0 & 6 & 3 & 3 & -6 \\ -1 & 3 & 4 & 2 & -5 \\ 1 & 3 & 2 & 4 & -4 \\ 2 & -6 & -5 & -4 & 10 \end{pmatrix}$	15	108	162	96	15	3
6	$\begin{pmatrix} 4 & 0 & 0 & 0 & 2 \\ 0 & 4 & 2 & 2 & -4 \\ 0 & 2 & 4 & 0 & -2 \\ 0 & 2 & 0 & 4 & -4 \\ 2 & -4 & -2 & -4 & 8 \end{pmatrix}$	15	128	192	96	15	3
7	$\begin{pmatrix} 4 & -2 & 0 & 0 & 2 \\ -2 & 4 & 0 & 0 & -2 \\ 0 & 0 & 4 & 2 & -2 \\ 0 & 0 & 2 & 4 & -2 \\ 2 & -2 & -2 & -2 & 4 \end{pmatrix}$	15	144	192	144	15	2
8	$\begin{pmatrix} 6 & -3 & -3 & -3 & 9 \\ -3 & 4 & 2 & 1 & -6 \\ -3 & 2 & 4 & 2 & -6 \\ -3 & 1 & 2 & 4 & -6 \\ 9 & -6 & -6 & -6 & 18 \end{pmatrix}$	15	81	162	144	15	2
9	$\begin{pmatrix} 4 & -2 & -2 & -2 & 6 \\ -2 & 4 & 2 & 2 & -6 \\ -2 & 2 & 4 & 2 & -4 \\ -2 & 2 & 2 & 4 & -6 \\ 6 & -6 & -4 & -6 & 16 \end{pmatrix}$	15	80	192	240	15	2

TABLE 5.1. The ten vertices J_i of $\mathcal{R}$ up to equivalence. $\lvert\operatorname{Min}(J_i)\rvert$ counts pairs $\pm k$; Γ_{J_i} is the stabiliser of J_i in Γ; "rays" is the number of extreme rays of C_{J_i} and "orbits" the number of their orbits under Γ_{J_i}. The pairs $(\det Q, \det J)$ are distinct, so no two J_i are equivalent.

(1) *for every i and every extreme ray Y of C_{J_i} with $t^*(J_i, Y) < \infty$, the vertex $J_i + t^*Y$ is equivalent to some J_j, and $(i, j) \neq (0, 0)$;*
(2) *every extreme ray Y of C_{J_i} with $t^*(J_i, Y) = \infty$ equals $X_{u,a}$ (Definition 5.2) for some $u \in \mathbb{Z}^4$, $a \in \mathbb{Z}$, and is equivalent under Γ_{J_i} to one of the seven rays in Table 5.3.*

Proof. This is a finite computation in exact integer and rational arithmetic, described in Appendix A. For each of the 1112 bounded edges it produces t^*, the class j and a matrix $T \in \Gamma$ with $T^{\mathsf{T}} J_j T = J_i + t^* Y$; for each of the 25 unbounded edges it produces (u, a). These certificates and the completeness of the lists of extreme rays are checked by independent programs. □

i	end points of the edges at J_i: class j of $J_i + t^*Y$, with t^*, and number of edges (orbits)
0	1 (2): 16 (1); 2 (2): 24 (1); 3 (2): 16 (1); ∞: 4 (1)
1	0 (2): 4 (1); 1 (2): 36 (2); 2 (2): 24 (1); 3 (2): 64 (2); 4 (2): 12 (1); 5 (1): 12 (1); 6 (2): 12 (1); ∞: 5 (2)
2	0 (2): 3 (1); 1 (2): 12 (1); 3 (2): 12 (1); 7 (2): 4 (1); ∞: 2 (1)
3	0 (2): 5 (1); 1 (2): 80 (2); 2 (2): 30 (1); 3 (2): 155 (4); 4 (2): 50 (2); 5 (1): 20 (1); 6 (2): 20 (1); 7 (2): 15 (1); 8 (1): 15 (1); 9 (2): 5 (1); ∞: 5 (1)
4	1 (2): 48 (1); 3 (2): 160 (2); 4 (2): 112 (2); 5 (1): 8 (1); 8 (1): 32 (1); 9 (2): 32 (1); ∞: 8 (1)
5	1 (1): 6 (1); 3 (1): 8 (1); 4 (1): 1 (1)
6	1 (2): 6 (1); 3 (2): 8 (1); ∞: 1 (1)
7	2 (2): 6 (1); 3 (2): 9 (1)
8	3 (1): 9 (1); 4 (1): 6 (1)
9	3 (2): 5 (1); 4 (2): 10 (1)

TABLE 5.2. The edges at the vertices J_i. An entry $j\,(t^*)\colon N\,(o)$ means that N extreme rays Y of C_{J_i}, forming o orbits under Γ_{J_i}, have $t^*(J_i, Y) = t^*$ and $J_i + t^*Y$ equivalent to J_j; the entry ∞ counts the unbounded edges. No edge joins two vertices of class 0.

By Corollary 2.11 and Proposition 5.1(1), every vertex of $\mathcal{R}$ is equivalent to exactly one J_i, and every edge of $\mathcal{R}$ is equivalent to an edge at some J_i.

Definition 5.2. For $u \in \mathbb{Z}^4$ and $a \in \mathbb{Z}$ let $p = (u, -a)$, $q = (u, -a-1) \in \mathbb{Z}^5$ and $X_{u,a} = pq^{\mathsf{T}} + qp^{\mathsf{T}}$. Thus $Q(X_{u,a}) = 2uu^{\mathsf{T}}$ and

$$X_{u,a}[(n,l)] = 2\,(u\cdot n - al)\,(u\cdot n - (a+1)l).$$

Lemma 5.3. *$X_{u,a}[k] \geq 0$ for all $k \in \mathcal{M}$. Hence $J + tX_{u,a} \in \mathcal{R}$ for all $J \in \mathcal{R}$ and $t \geq 0$, and*

$$\det Q(J + tX_{u,a}) = \det Q(J) + 2t\ \mathrm{adj}(Q(J))[u].$$

Proof. For $k = (n, l)$ the two factors $u\cdot n - al$ and $u\cdot n - (a+1)l$ are integers that differ by $l \in \{-1, 0, 1\}$; two such integers have a nonnegative product. The determinant formula is the matrix determinant lemma for the rank-one update $Q + 2t\,uu^{\mathsf{T}}$. □

Proposition 5.4. *Every $J \in \mathcal{C}$ is equivalent to J_0 or to $J_i + t_0X_{u,a}$ for a row of Table 5.3 with a root t_0. In particular $\det Q(J) \geq 256$ for all $J \in \mathcal{C}$, and $\det Q(J) = 256$ only if J is equivalent to J_0 or to $J_1 + 2X_{e_1,-1}$.*

Proof. Let $J \in \mathcal{C}$. If $d(J) = 0$, then J is equivalent to some J_i, and $\sigma(J) = \sigma(J_i) = \det J_i / \det Q(J_i)$. By Table 5.1 this is positive unless $i = 0$, and $J \in S$ requires $\sigma(J) \leq 0$; so J is equivalent to J_0.

If $d(J) = 1$ and $\sigma(J) = 0$, Lemma 2.9(2) places J in the relative interior of an edge, and after applying an element of Γ we may assume $J = J_i + tY$ with Y an extreme ray of C_{J_i} and $0 < t < t^*(J_i, Y)$. Suppose $t^* < \infty$. By Proposition 5.1(1) the end point $J_i + t^*Y$ is equivalent to some J_j with $(i,j) \neq (0,0)$, so $\sigma \geq 0$ at both end points and $\sigma > 0$ at one of them. The segment lies in $\mathcal{R} \subset \mathrm{Sym}_5^+$, where σ is concave (Lemma 2.3(2)), so

$$\sigma(J_i + tY) \ \geq\ \Big(1 - \frac{t}{t^*}\Big)\sigma(J_i) + \frac{t}{t^*}\,\sigma(J_i + t^*Y) \ >\ 0,$$

i	(u,a)	N	$\det Q(J_i+tX_{u,a})$	$\det(J_i+tX_{u,a})$	$t_0>0$	$\det Q$ at t_0
0	$(e_1,-1)$	4	$128(t+2)$	$-64\,t(t+2)$	none	—
1	$((0,1,1,1),1)$	4	$32(3t+4)$	$-16(t-2)(3t+4)$	2	320
1	$(e_1,-1)$	1	$64(t+2)$	$-32(t-2)(t+2)$	2	256
2	$(e_1,-1)$	2	$96(t+2)$	$-16(t+2)(3t-4)$	4/3	320
3	$((1,-1,-1,-1),-3)$	5	$16(4t+5)$	$-32(t^2-2t-4)$	$1+\sqrt{5}$	$144+64\sqrt{5}$
4	$(e_3,0)$	8	$64(t+1)$	$-32(t-4)(t+1)$	4	320
6	$(e_1,-1)$	1	$64(t+2)$	$-32(t-3)(t+2)$	3	320

TABLE 5.3. The unbounded edges $J_i+tX_{u,a}$, $t\ge 0$, one per orbit under Γ_{J_i}; N is the size of the orbit. The last two columns give the positive root t_0 of $\det(J_i+tX_{u,a})$ and the value of $\det Q$ there. Each row is a direct computation (the $\det Q$ column by Lemma 5.3); $144+64\sqrt{5}\approx 287.11$.

a contradiction. Hence $t^*=\infty$. By Proposition 5.1(2), after applying an element of Γ_{J_i}, $Y=X_{u,a}$ for a row of Table 5.3, and $\sigma(J)=0$ means $\det(J_i+tX_{u,a})=0$ (Lemma 2.3(3)), so $t=t_0$. The values of $\det Q$ are read off Table 5.3 and Table 5.1. □

6. Proofs of Theorem 1.1 and Corollary 1.2

Proof of Theorem 1.1. Let Λ and b be as in the theorem, a a basis of Λ and $J=J(a,b)$. By Lemmas 2.1 and 2.4, $J\in\mathcal{R}$, $\sigma(J)=0$, so $J\in S$, and $\det Q(J)=\operatorname{covol}(\Lambda)^2$. Propositions 3.2 and 5.4 give $J'\in\mathcal{C}$ with $256\le\det Q(J')\le\det Q(J)$. Hence $\operatorname{covol}(\Lambda)\ge 16$. The same argument applied to arbitrary points of S shows $\inf_S\det Q\ge 256$.

The set $\sqrt{2}\,D_4=\sqrt{2}\,\Lambda_2\cup(\sqrt{2}\,\Lambda_2+\sqrt{2}\,e_1+\sqrt{2}\,e_3)$ satisfies the hypotheses with $\operatorname{covol}(\sqrt{2}\,\Lambda_2)=\sqrt{\det G_2}=16$.

Suppose $\operatorname{covol}(\Lambda)=16$. Then $\det Q(J)=256=\inf_S\det Q$, and Proposition 3.2 gives $J'\in\mathcal{C}$ with $Q(J')=Q(J)$. By Proposition 5.4, $J'=T^{\mathsf{T}}J''T$ with $T\in\Gamma$ and $J''\in\{J_0,\ J_1+2X_{e_1,-1}\}$, so $Q(J)=U^{\mathsf{T}}Q(J'')U$ with $U\in\mathrm{GL}_4(\mathbb{Z})$. Now $Q(J_0)=4I_4=G_2$ and

$$J_1+2X_{e_1,-1}=\begin{pmatrix}8&0&0&0&4\\0&4&2&2&-4\\0&2&4&2&-4\\0&2&2&4&-4\\4&-4&-4&-4&8\end{pmatrix},\qquad Q(J_1+2X_{e_1,-1})=G_1,$$

with G_1,G_2 as in (4.1). So Λ, whose Gram matrix in the basis aU^{-1} is G_1 or G_2, equals $\phi(\sqrt{2}\,\Lambda_1)$ or $\phi(\sqrt{2}\,\Lambda_2)$ for a linear isometry ϕ. Since $\operatorname{dist}(b,\Lambda)\ge 2$, Corollary 4.3 gives $P=\phi(\sqrt{2}\,D_4)$.

Finally, P is the union of the two disjoint translates Λ and $\Lambda+b$, so it has two points per fundamental domain of Λ, and the density of the packing of unit balls centred at P is $2\cdot\frac{\pi^2}{2}/\operatorname{covol}(\Lambda)=\pi^2/\operatorname{covol}(\Lambda)\le\pi^2/16$. □

Proof of Corollary 1.2. Let the centres be $P=\Lambda\cup(\Lambda+b)$, $b\notin\Lambda$; if $b\in\Lambda$ the centres form a lattice and the statement is the theorem of Korkine and Zolotarev [5]. The map $x\mapsto b-x$ maps Λ onto $\Lambda+b$ and $\Lambda+b$ onto Λ, so it is an isometry of P; it maps the Voronoi cell V_0 of 0 onto the cell V_b of b. Translations by Λ permute the cells, and $V_0\cup V_b$ is a fundamental

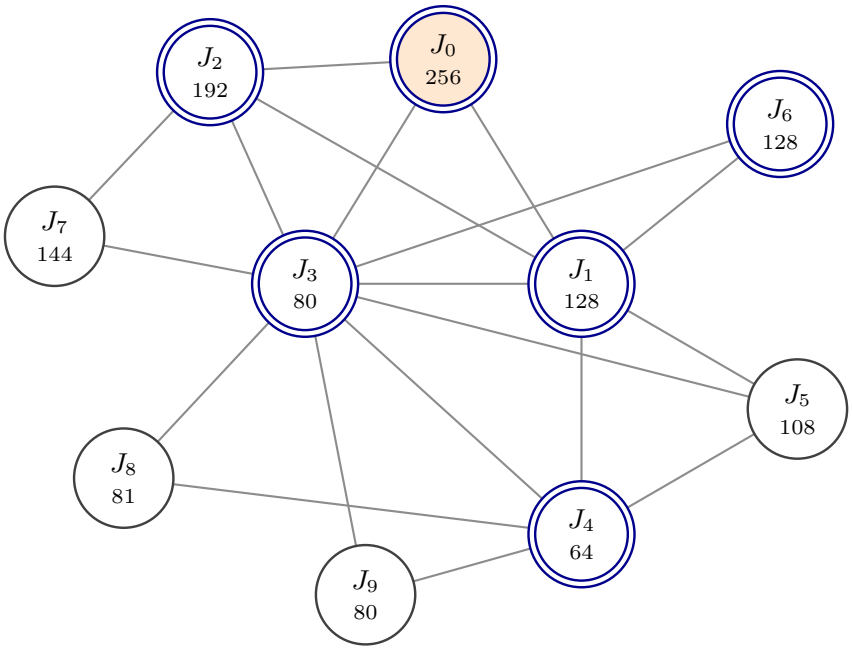


FIGURE 6.1. The ten classes of vertices of $\mathcal{R}$, labelled with $\det Q$, and the edges between different classes (Table 5.2). The classes of J_1, J_3 and J_4 also have edges joining two vertices of the same class. Double circles mark the classes with unbounded edges. The packings of maximal density are the vertex J_0 and a point on an unbounded edge at J_1.

domain of Λ up to a null set. Hence every cell has volume $\frac{1}{2}\operatorname{covol}(\Lambda) \geq 8$ by Theorem 1.1, with equality only for $P \cong \sqrt{2}\, D_4$, whose Voronoi cell is the regular 24-cell [4, Ch. 21]. □

APPENDIX A. COMPUTATION AND VERIFICATION

The computation behind Proposition 5.1 follows [1] and Voronoi's algorithm [8, 7], in exact arithmetic throughout. All files are archived with the paper [2]; the script `verification/shell/run_all.sh` runs every check. The author used Claude (Anthropic), an AI system, for programming, computation and drafting; the author checked the mathematics and is responsible for the content.

Enumeration. Starting from J_0, for each new vertex J the program `ryshkov2.py` (Python, rational arithmetic) computes $\operatorname{Min}(J)$ by the enumeration method of Fincke and Pohst, applied to the forms of Lemma 2.3(1) with rational bounds for the square roots; computes the extreme rays of C_J with `cddlib` in GMP rational arithmetic; for each ray Y decides whether $t^* = \infty$ and otherwise finds t^* exactly (by the sign pattern of $Q(Y)$, a unimodular reduction of its kernel, and a descent through crossing points $t = (J[k] - 4)/(-Y[k])$); and tests Γ-equivalence of $J + t^*Y$ with the known vertices by a backtracking search over images of minimal vectors. The search closes after ten classes and 1137 rays, and `certify.py` writes the certificates of Proposition 5.1 as integer tables.

Independent checks.

(1) **C** (`check_polyhedron.c`, integers with overflow traps): recomputes $\mathrm{Min}(J_i)$ and checks $\min_{\mathcal{M}^*} J_i = 4$ and rank 15; recomputes all extreme rays of each C_{J_i} from all 14-element subsets of $\mathrm{Min}(J_i)$, independently of `cddlib`, and compares them with the list; for every bounded edge checks $\min_{\mathcal{M}^*}(J_i + t^*Y) = 4$, rank 15, $T \in \Gamma$ and $T^{\mathsf{T}} J_j T = J_i + t^*Y$; for every unbounded edge checks $Y = X_{u,a}$.

(2) **Julia** (`check_candidates.jl`, rational arithmetic): without using the concavity argument of Proposition 5.4, isolates with Sturm sequences all roots of $\det(J_i + tY)$ in $(0, t^*)$ on all 1137 edges, decides the sign of $\det Q - 256$ at each, and checks that the points with $\det Q = 256$ are J_0 and the point $J_1 + 2X_{e_1,-1}$ (up to the orbit). It finds 21 edge points, as in Table 5.3.

(3) **Lean 4** (`R4Check.lean`): proves the integer inequality of Lemma 5.3 in the kernel, and checks by evaluation the invariants of Table 5.1, the minimal vectors, the 1112 relations $T^{\mathsf{T}} J_j T = J_i + t^*Y$ with $T \in \Gamma$, the 25 identities $Y = X_{u,a}$, and the orbit decomposition of the unbounded edges in Table 5.3.

(4) **Python** (`edges.py`, `sympy`): repeats the candidate analysis of the Julia check with exact real-root isolation.

Floating-point numbers appear only in the C program, to choose enumeration boxes, which are widened by two units in every coordinate; every decision is made in integer arithmetic. The Gram determinants $144 + 64\sqrt{5}$ and 320 of the non-maximal candidates agree with the densities $2/\sqrt{144 + 64\sqrt{5}}$ and $1/(4\sqrt{5})$ reported in [1, §4.2].

Statements and Declarations

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Competing interests. The author has no competing interests to declare that are relevant to the content of this article.

Data and code availability. The code, the data and the certificates of Appendix A are available at https://github.com/creelie/eD424 and are archived on Zenodo [2].

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